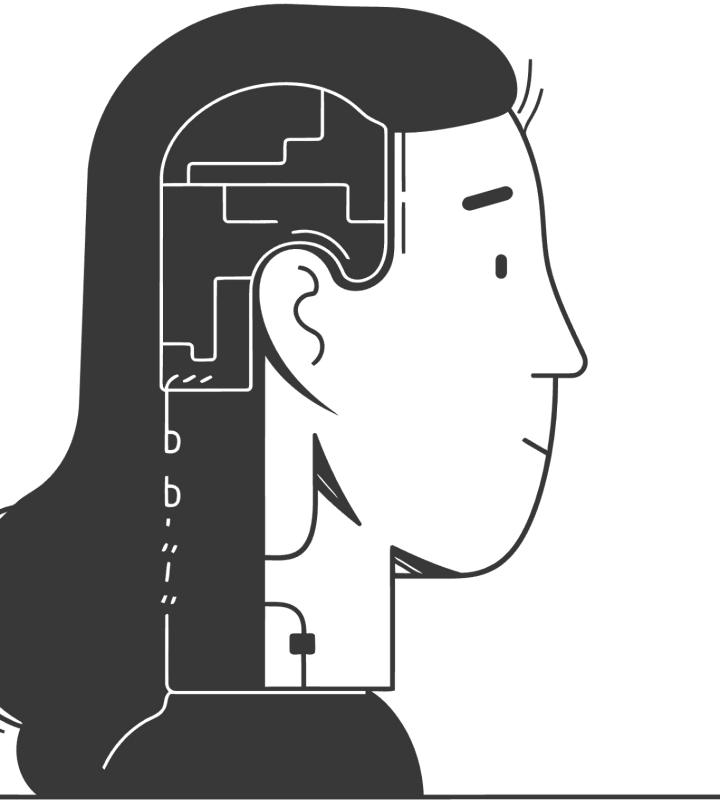


The DATA-MD Curriculum

Session 1 – Introduction to Artificial Intelligence

Data-Driven, Technology-Assisted Medical Decision Making for Medical Trainees

Emily Balczewski, PhD · Andrew Wong, MD, MS · Cornelius James, MD · Niharika Goparaju, MD · Thomas East, PhD · Joanne Baker, DO · Thalia Nguyen, MD · Serena Wang, MD · Lisa Zickuhr, MD

Pre-Test Assessment

Before we begin — let's benchmark your current understanding of AI in medicine.



Scan the QR code to complete the pre-test assessment. Your responses are anonymous and help us tailor the session.

NOW ENTERING

Session 1 — Introduction to AI

Establishing the foundation: what AI is, why clinicians must understand it, and where it's already reshaping medicine.



Understand AI Basics

Learn core concepts and terminology

Recognize Clinical Applications

Identify AI uses in diagnostics and care

Develop Critical Awareness

Assess risks, biases, and ethical issues

COMING NEXT

Session 2 — Error & Bias in AI Systems

Understanding where AI goes wrong — and why that matters for patient safety and equitable care.

- Sources of Bias
- Types of Error
- Prompting (Practical Application)

COMING NEXT

Sessions 3 & 4 — Critical Appraisal of AI Literature

Session 3 · Critical Appraisal + Evaluation

- AI benchmarks explained
- Reading the literature critically
- Frameworks for real-world validity

Session 4 · Critical Appraisal + Future

- AI for clinical documentation
- Ambient scribes in practice
- Future directions of AI in medicine



Our Philosophy

Understand How AI Works

Beyond surface-level familiarity — know enough to ask the right questions

Use AI in Everyday Practice

Practical fluency with tools already in clinical environments

Critically Appraise the Literature

Evaluate AI studies with rigor and clinical relevance

...to be **effective, ethical, and satisfied** stewards of clinical AI for your patients and peers.

Session 1 Learning Objectives



AI Trends

Describe AI trends in medicine



Key Terms

Define foundational AI terminology



Model Lifecycle

Understand basic AI model development



LLM Training

Explain large language model training techniques



Ethics

Discuss general ethical considerations around AI use

KEY CONCEPT

Why Should We Care About AI in Healthcare?

Already Here

AI tools are embedded in EHRs, imaging workflows, and ICUs today

Rapidly Expanding

Investment and adoption are accelerating across every specialty

Affects Your Patients

Clinicians who understand AI can advocate for safer, more equitable care



AI Tools Already in Clinical Use



Medical Devices

Smart insulin pumps with adaptive dosing algorithms



Surgical Devices

Robotic surgery systems with AI-assisted precision



Imaging Readers

Automated CXR and EKG preliminary interpretations



Decision Support

Epic Sepsis Model and early warning systems



Documentation

Ambient scribes capturing clinical encounters in real time



Operations

ICU bed prioritization and resource allocation (often unregulated)

"85% of healthcare leaders globally say their organization is currently or planning to invest in generative AI technologies."

— Philips Health Index 2024

85%

Planning to Invest

Currently or planning to invest in generative AI

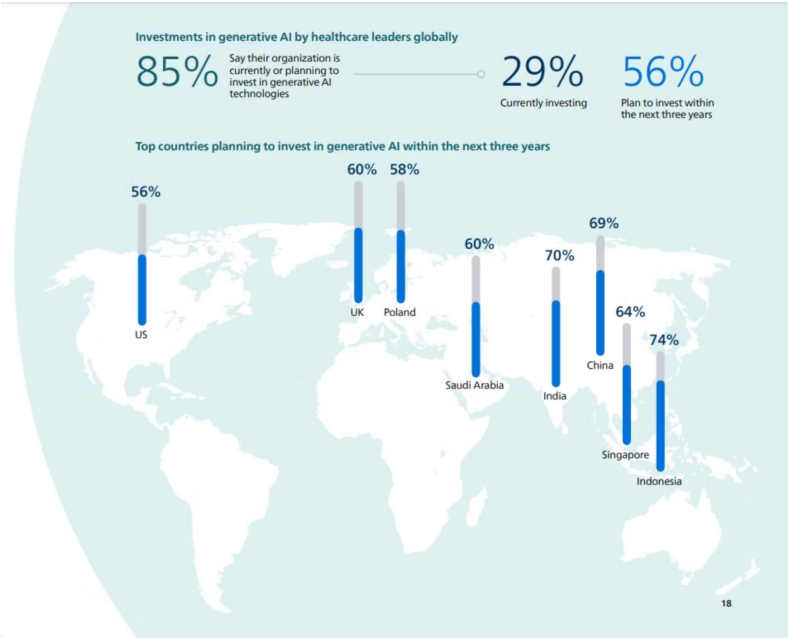
29%

Currently Investing

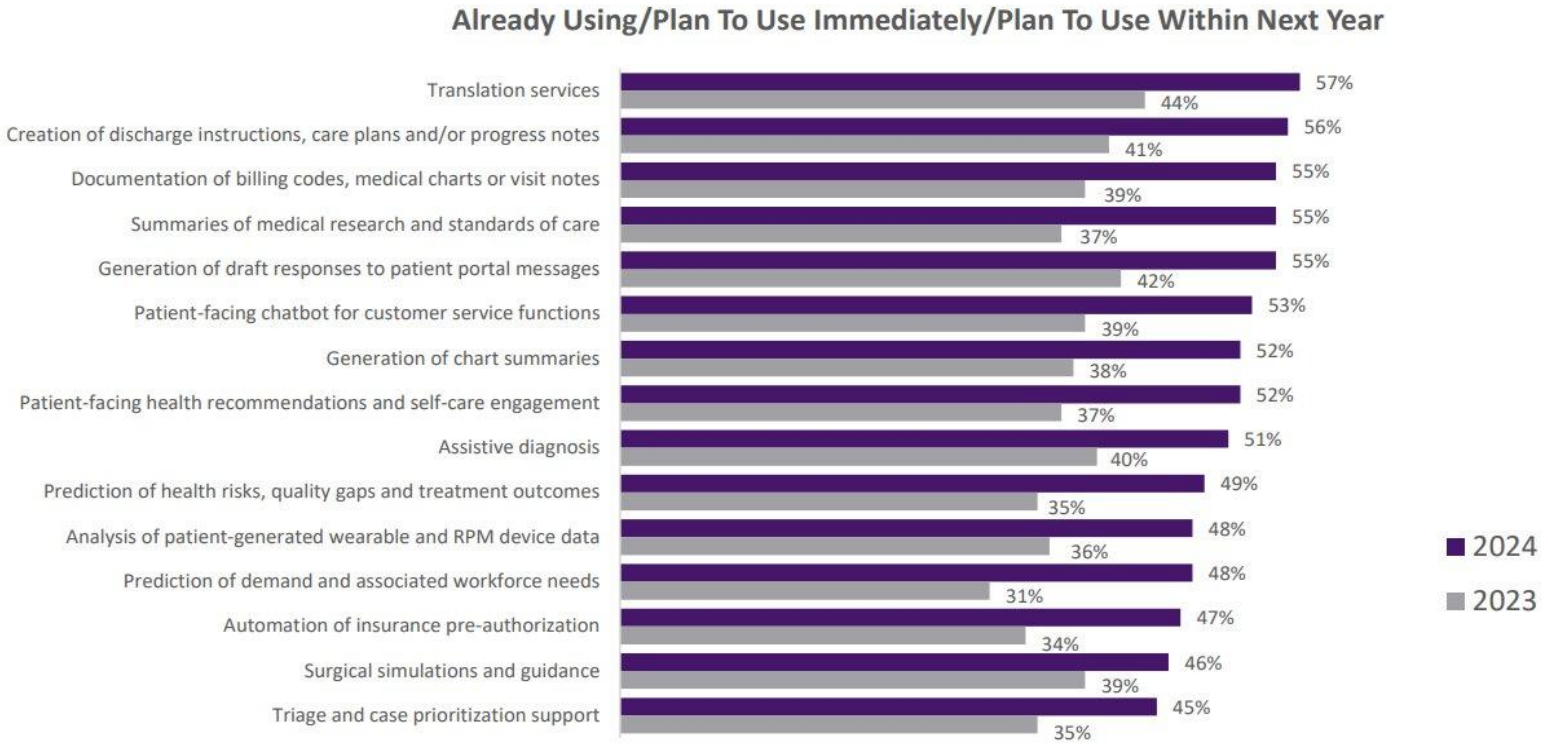
56%

Next 3 Years

Plan to invest within the next three years



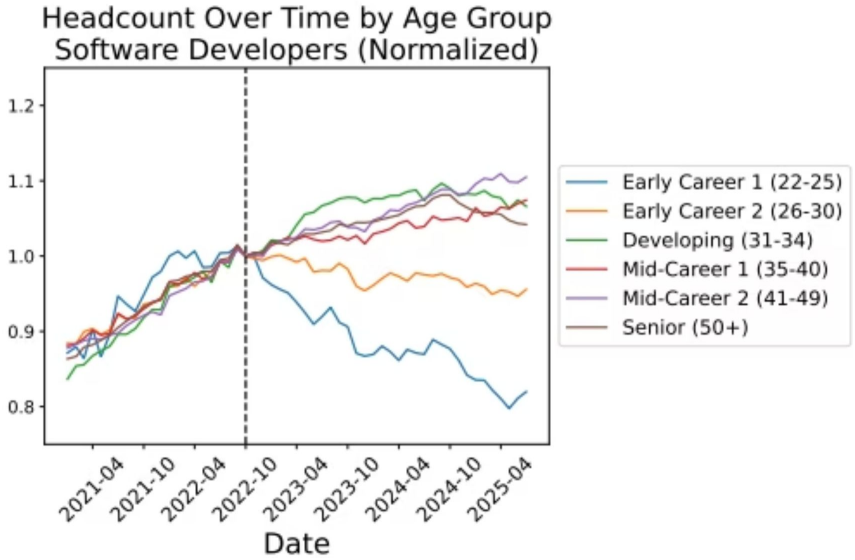
There was a significant increase in physicians already using or planning to use AI in the near term

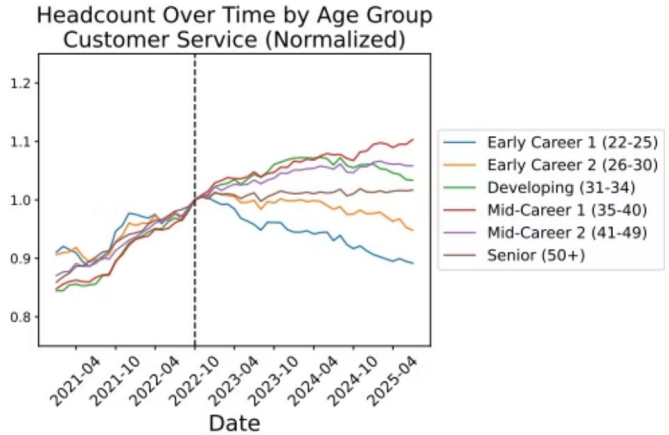


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When would you expect to start incorporating this AI use case into your own practice?

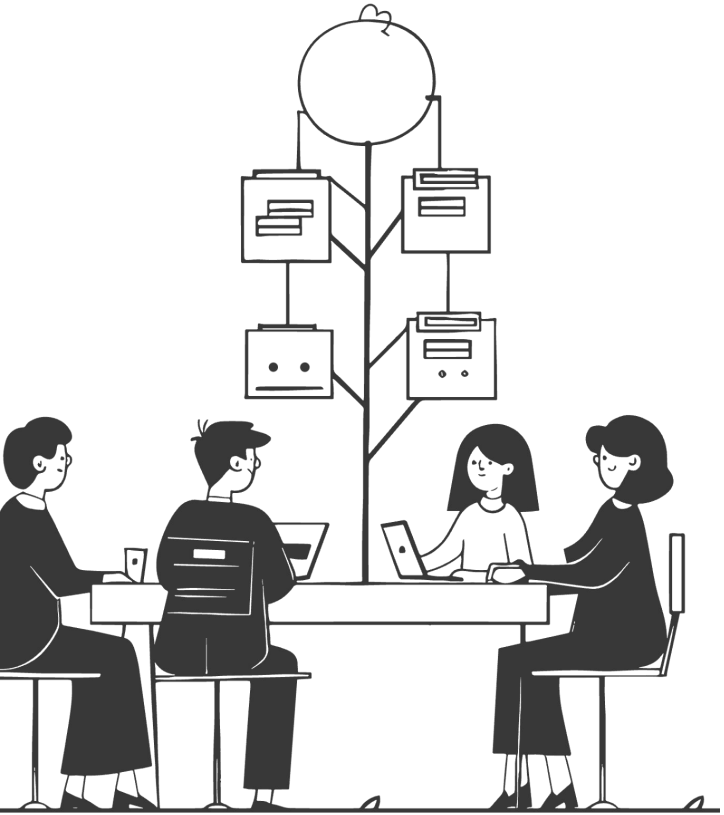
AI Revolution Will Disproportionately Affect Younger Workers — Including New Physicians





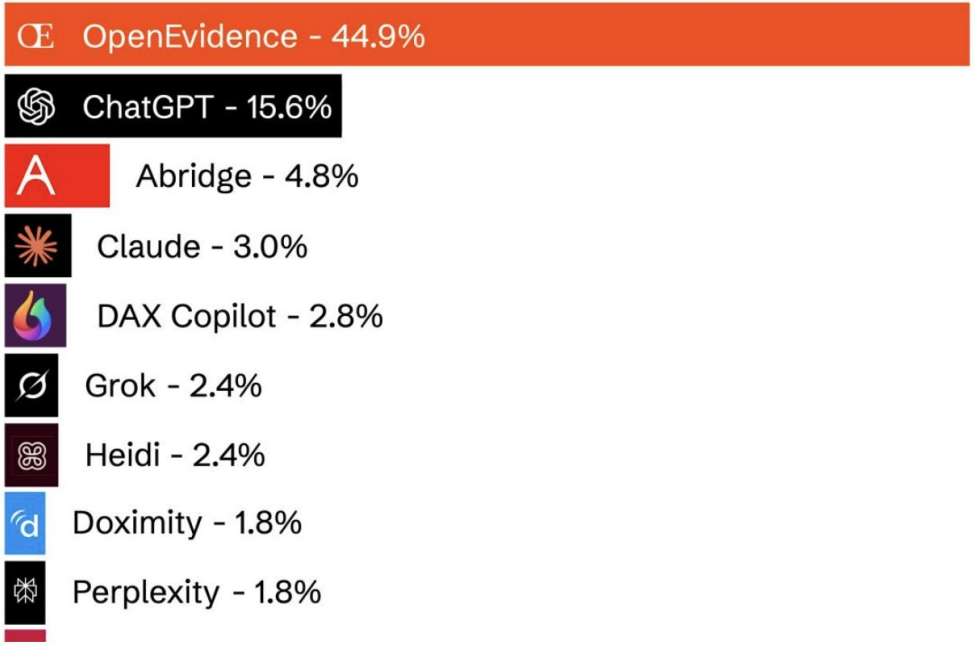
Early-career workers in software development and customer service show declining headcount post-2022.

Brynjolfsson et al., Stanford Digital Economy Lab (2025)



DISCUSSION QUESTION

Which AI Tools Have You Personally Used for Clinical or Professional Tasks?



Usage Level	Percentage
Daily (multiple times)	51%
Daily (once or twice)	16%
Sometimes (few times per week)	22%
Rarely (few times per month or less)	7%
Never	3%
Not sure/Don't know if I am	1%

At 44.9%, OpenEvidence is used by more American physicians than all other AI tools for physicians combined—including ChatGPT.

The 2025 Physician AI Report was published by Dr. Graham Walker's *Offcall*. The Report surveyed over 1,000 practicing U.S. physicians across specialties, ages, and career stages.

Offcall

The Adoption Paradox

Yet, despite the daily usage, **81%** of physicians say they are dissatisfied with how their organization is deploying it.



Offcall

Employer AI Adoption Speed Ratings



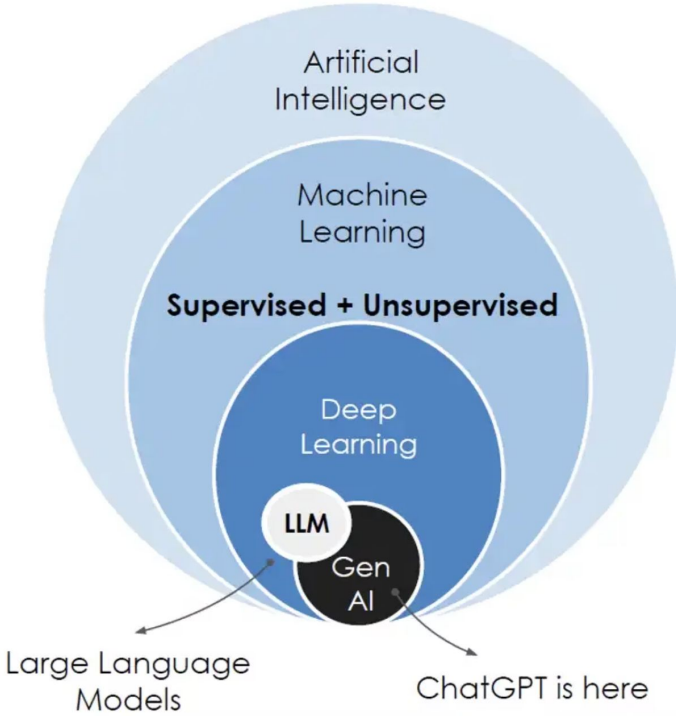
Response	Percentage
Just so-so, making some progress but not fast enough	42%
Not good, we are not adopting AI nearly fast enough	39%
Doing great and moving fast, I'm happy!	19%

Why are so many physicians dissatisfied? It comes down to three factors: (1) *Speed* (2) *Influence* (3) *Communication*.

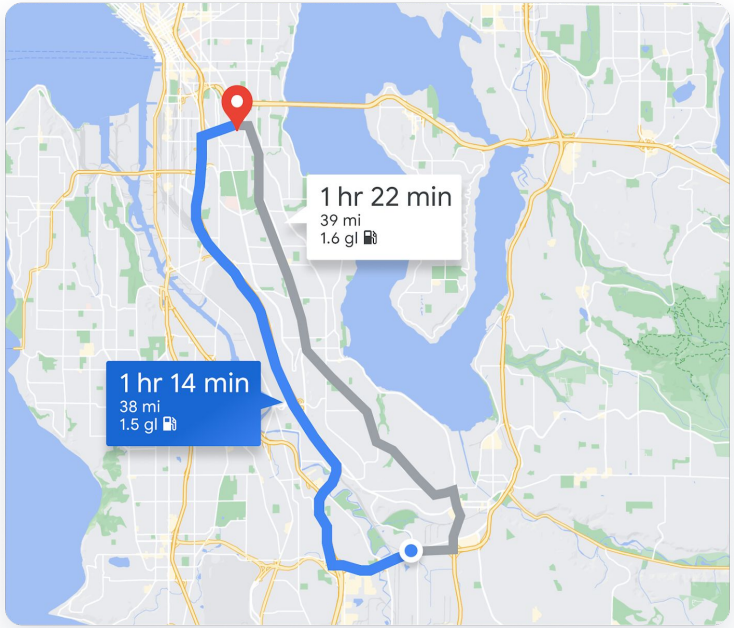
Offcall

KEY CONCEPT

What Is Artificial Intelligence?



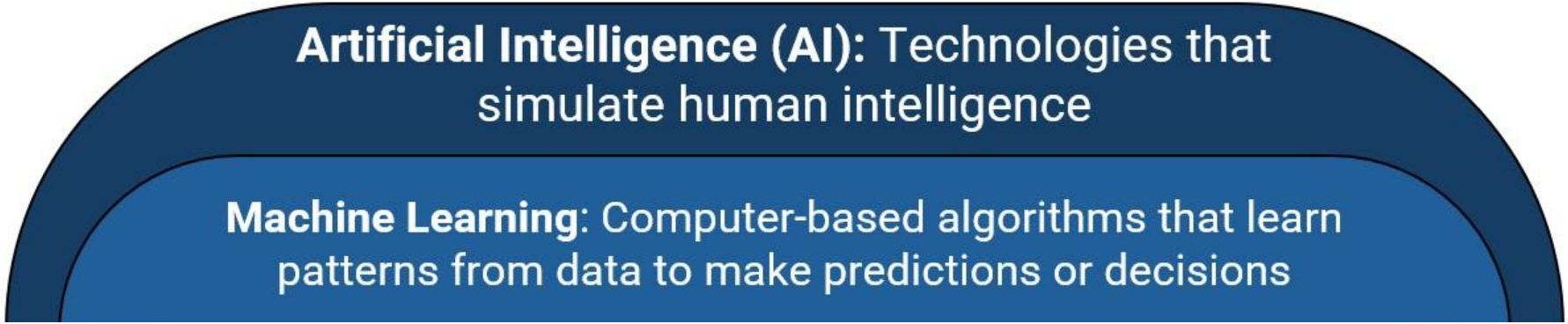
Artificial Intelligence (AI): Technologies that simulate human intelligence



maps.google.com



freethink.com



Classical Machine Learning **vs.** **Deep Learning**

Linear Regression *Neural Network*

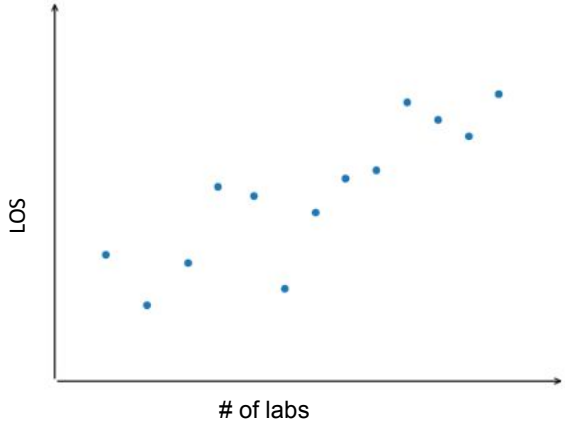
Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

Predict ED LOS

1) Gather data
2) Train the model
3) Implement the model

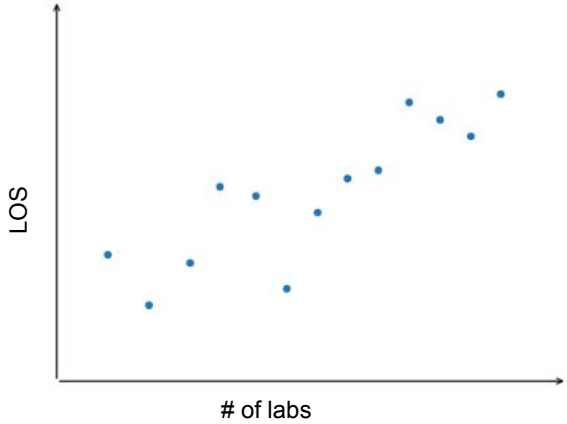


Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression
Predict ED LOS

1) Gather data
2) Train the model
3) Implement the model



Artificial Intelligence (AI): Technologies that simulate human intelligence

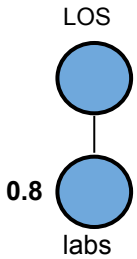
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

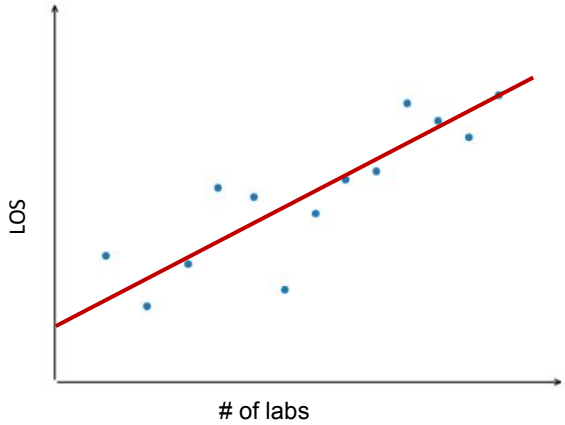
Linear Regression

Predict ED LOS

y = mx

LOS = m*labs





Artificial Intelligence (AI): Technologies that simulate human intelligence

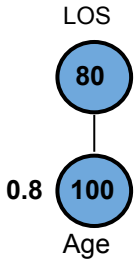
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

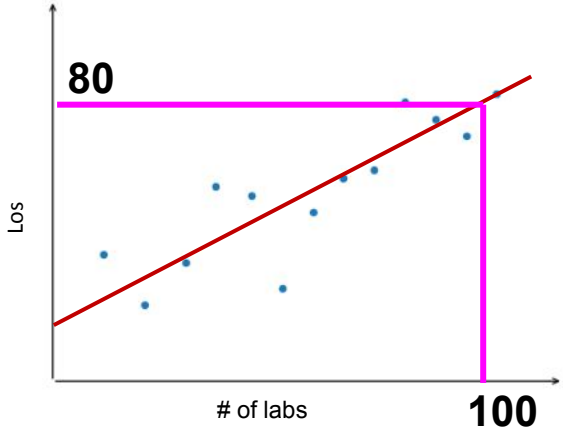
Predict ED LOS

y = m*x

LOS = 0.8*(# of Labs)



1) Gather data
2) Train the model
3) Implement the model



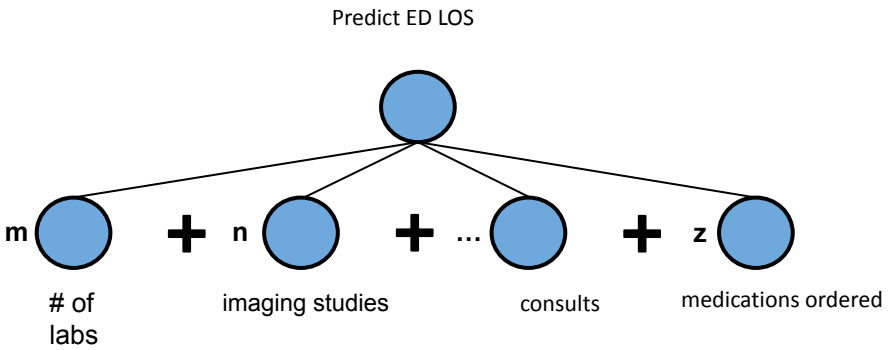
Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

Predict ED LOS

+ labs + … + consults



slope =
coefficient =
relationship between inputs and outputs =
weights

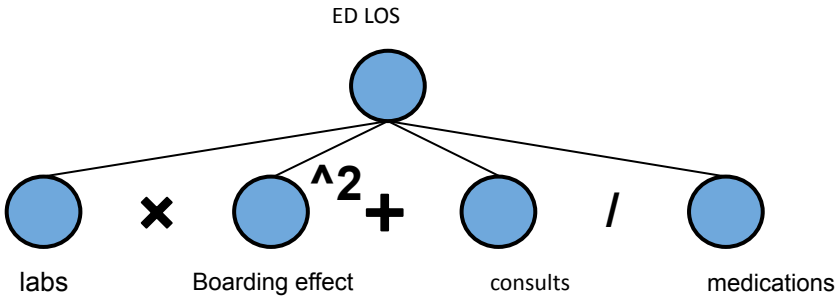
Artificial Intelligence (AI): Technologies that simulate human intelligence

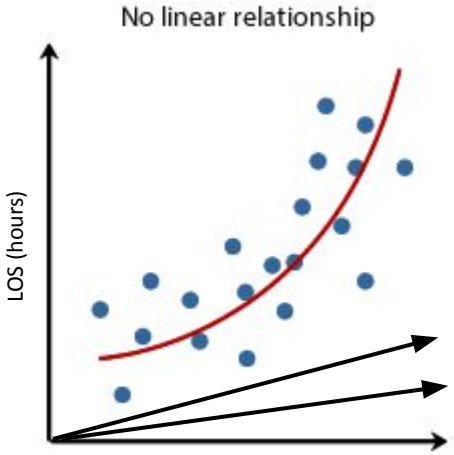
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Nonlinear Regression

Predict ED LOS

* Boarding effect^2 + … / consult

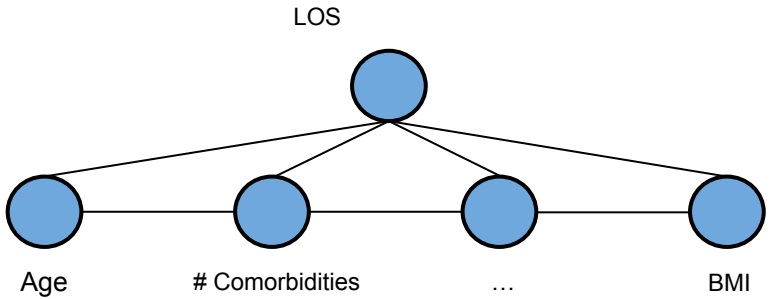




Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear/Nonlinear Regression
(Classical ML Models)



Pros:

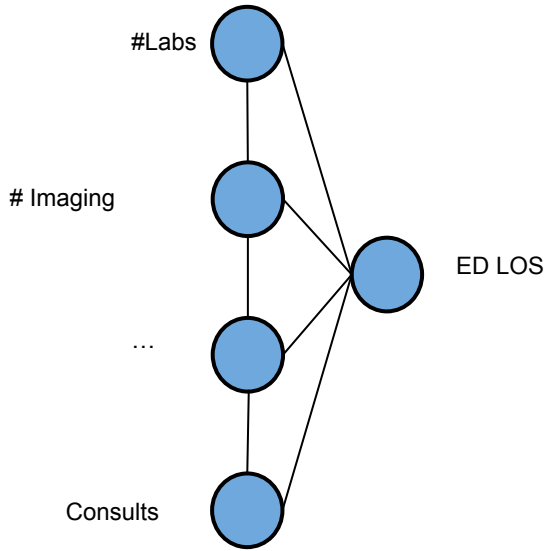
- Easy to understand; good transparency

Cons:

- You have to make assumptions about the kinds of relationships between variables
- May not work for modeling every kind of relationship

Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

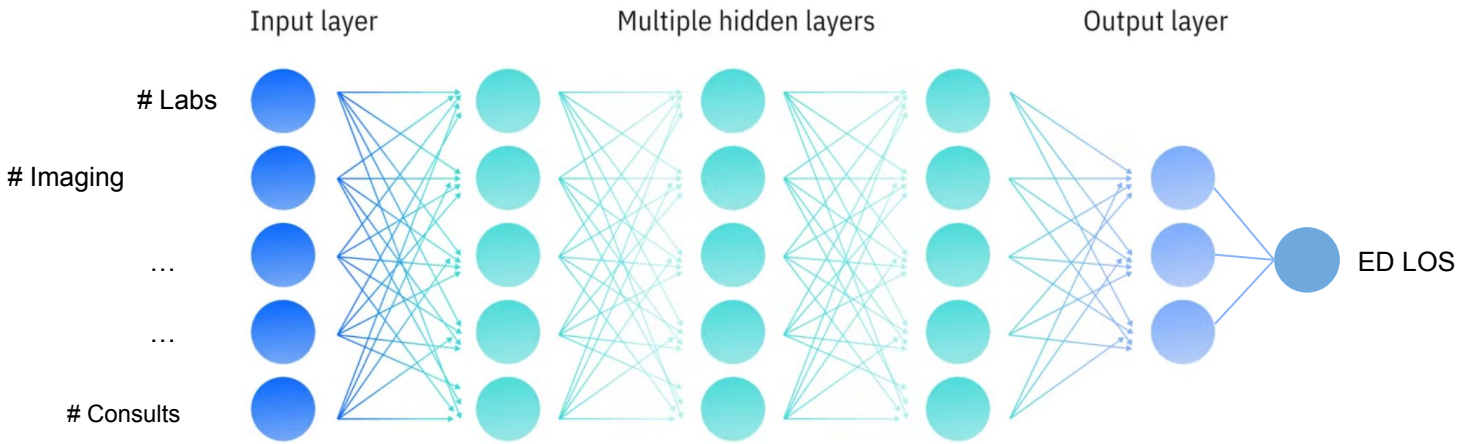


How do we get from Linear/Nonlinear Regression to Deep Learning?

Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Neural Network (Deep Learning)

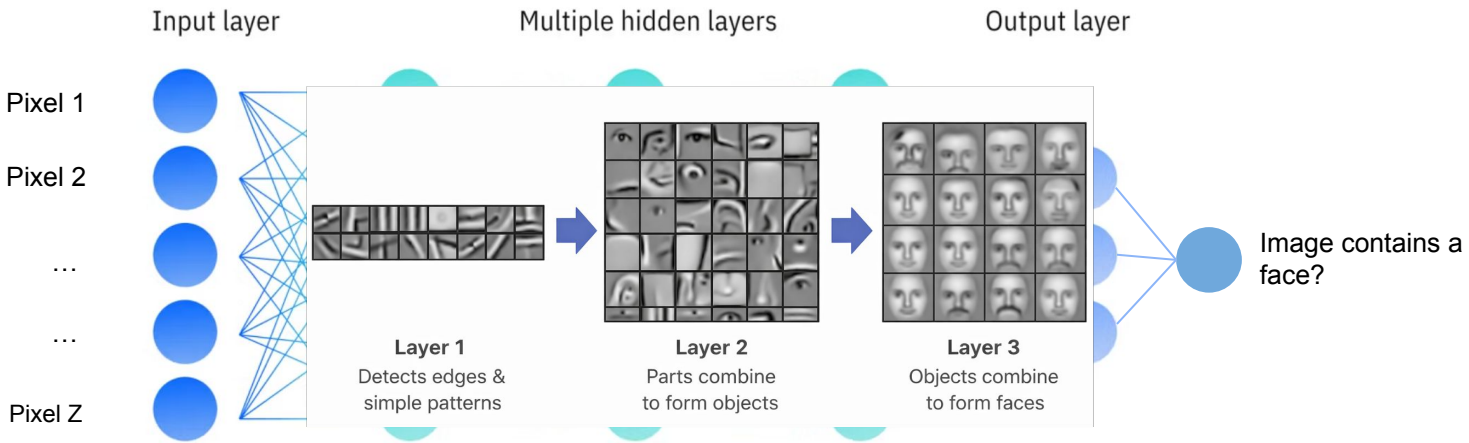


ibm.com

Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Neural Network (Deep Learning)

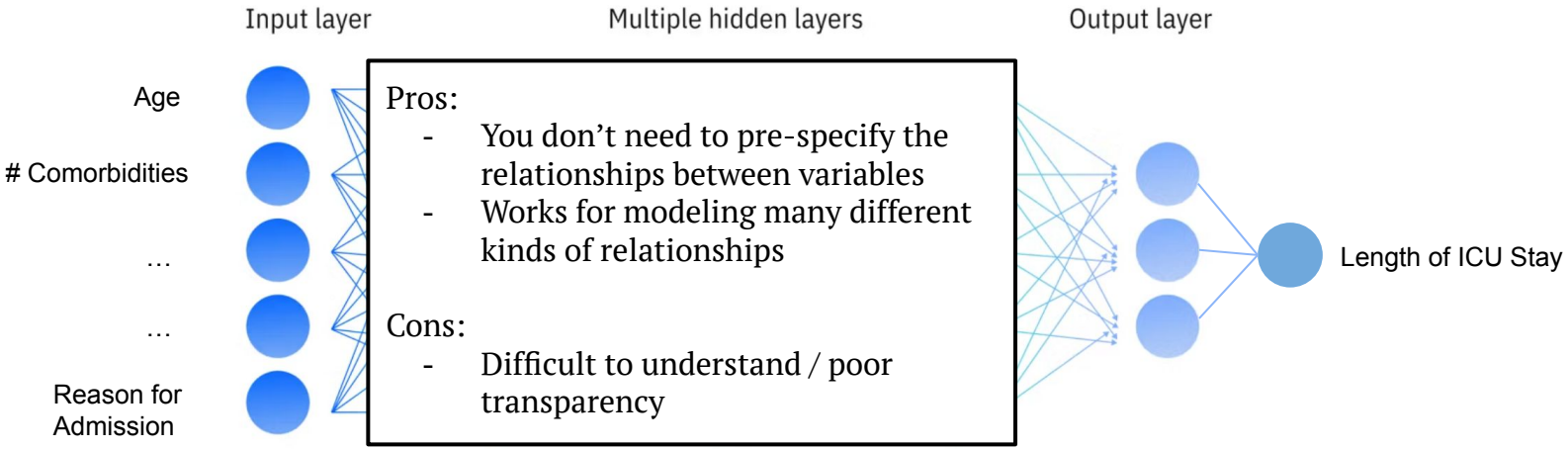


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Artificial Intelligence (AI): Technologies that simulate human intelligence

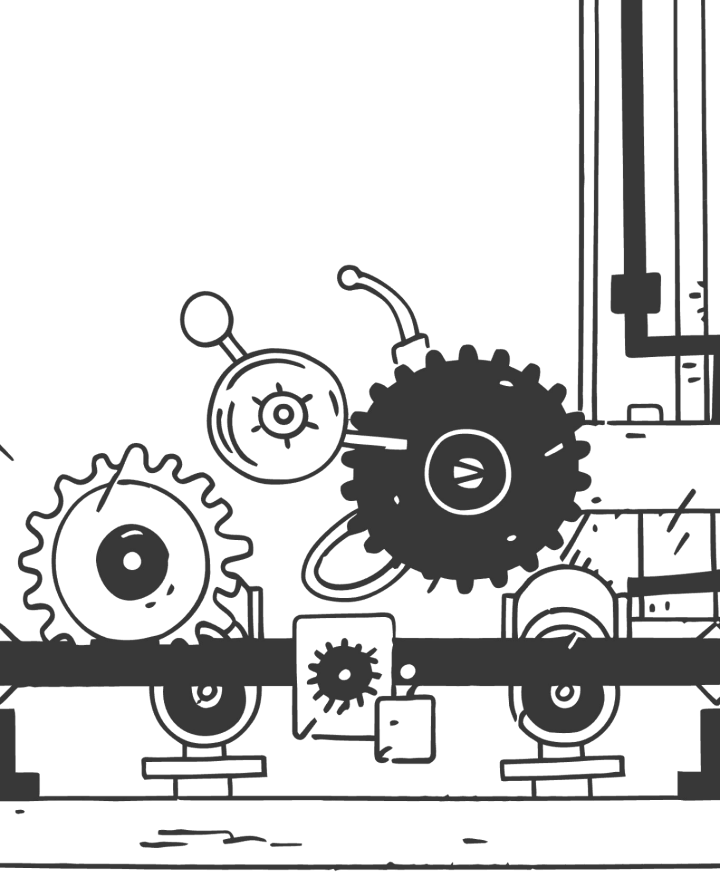
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Neural Network (Deep Learning)



Pros:
- You don't need to pre-specify the relationships between variables
- Works for modeling many different kinds of relationships

Cons:
- Difficult to understand / poor transparency



KEY CONCEPT

Deep Learning = Layered Linear Regression

1

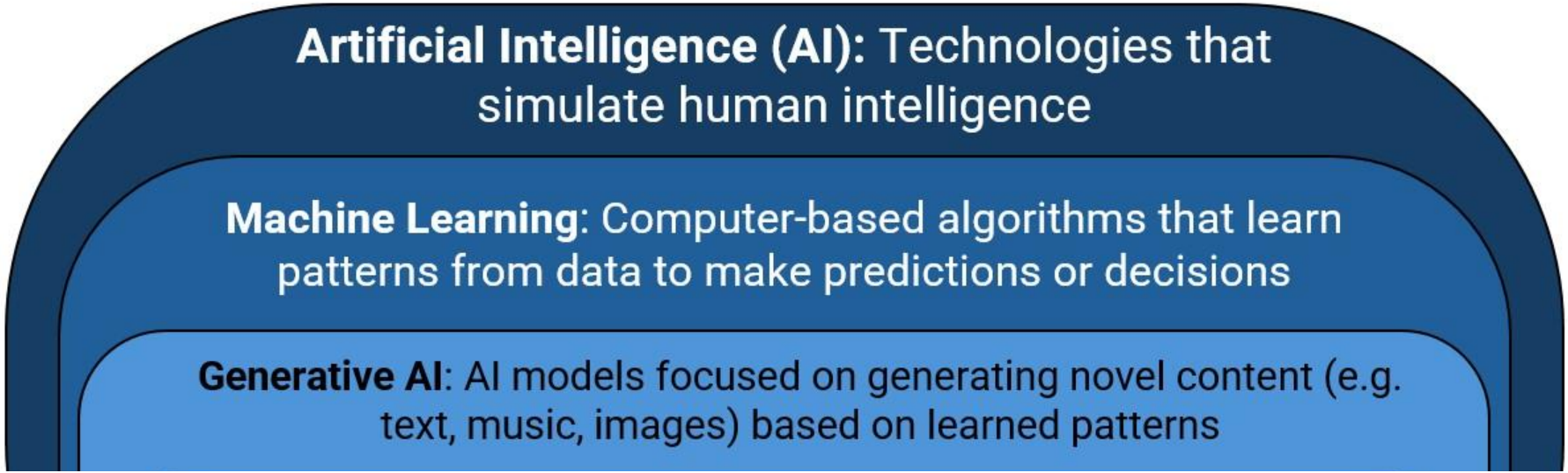
Linear Regression

1 layer, interpretable

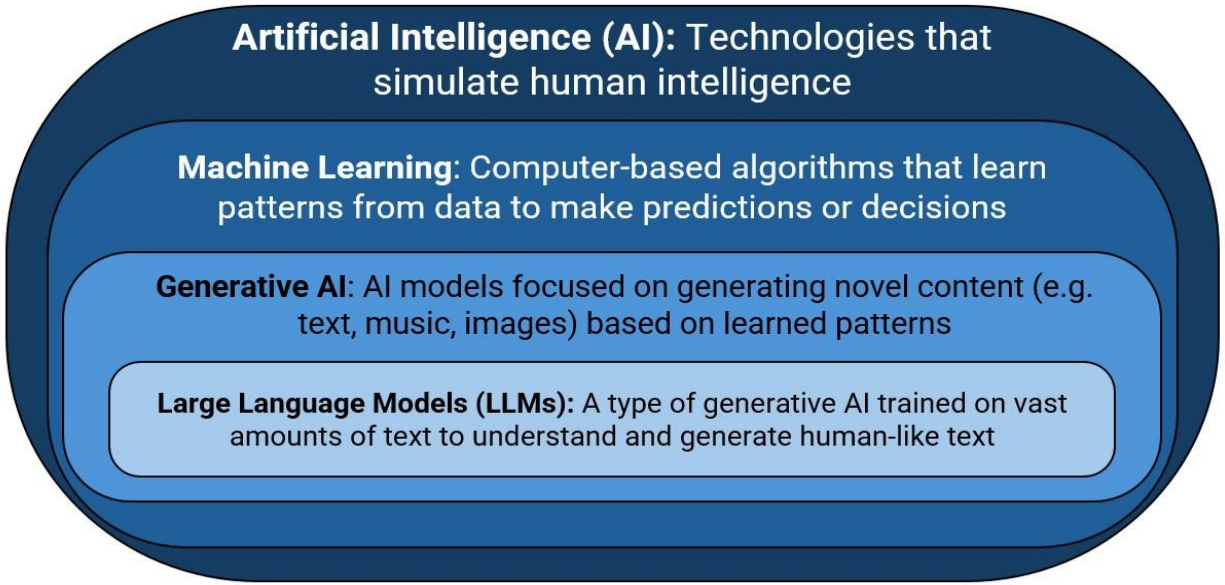
2

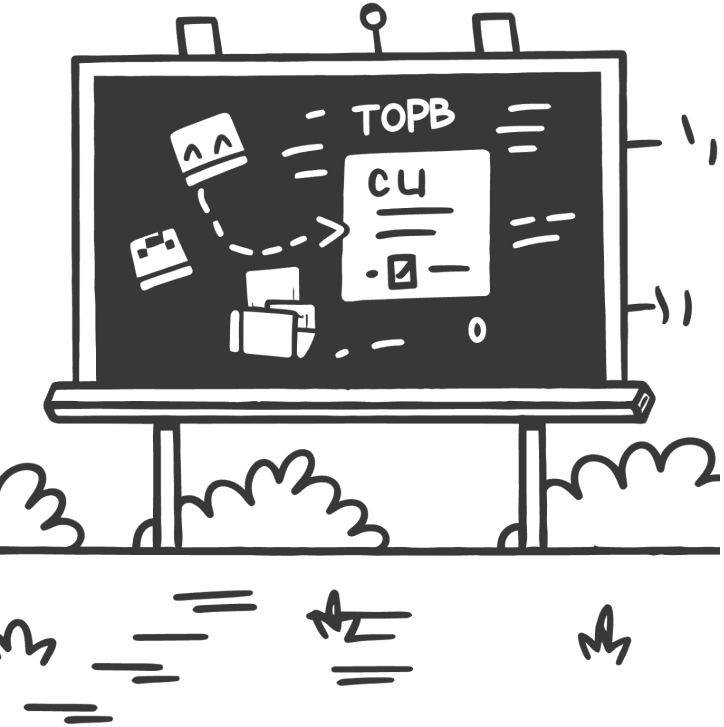
Deep Learning

Many layers, flexible, difficult to interpret


Artificial Intelligence (AI): Technologies that simulate human intelligence
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions
Generative AI: AI models focused on generating novel content (e.g. text, music, images) based on learned patterns

From Deep Learning to Large Language Models





LLMS DEFINED

What Makes LLMs Different?

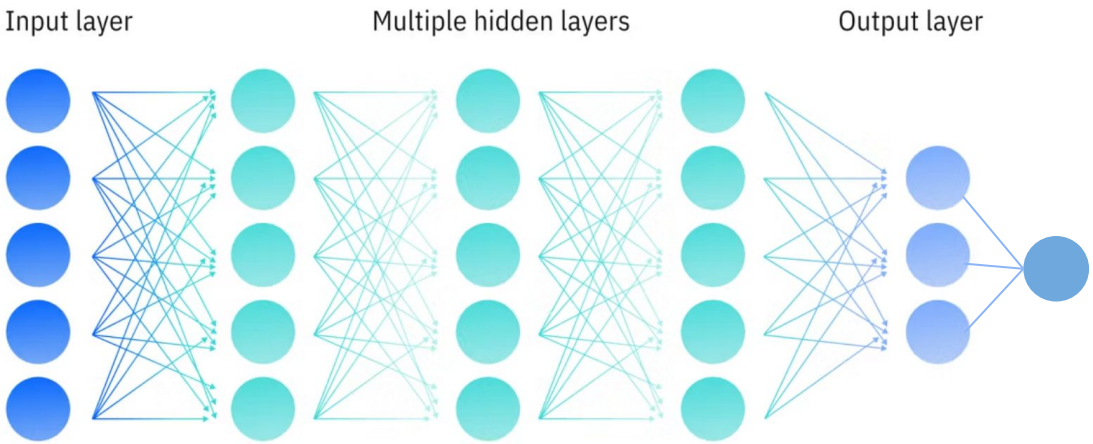
Trained on Vast Text

Books, articles, clinical notes, the web — billions of tokens

Generative

Produces novel text — not just predictions, but coherent language

Large Language Models (LLMs)



ChatGPT

OpenEvidence



LLAMA 3



LLM INPUT EXAMPLE

> "Complete this sentence: The patient with Type II Diabetes needs"

This is a **prompt** — the raw text input fed into the model. The LLM will process it token by token to generate a response.

Step 1: Tokenize: Sentence into words

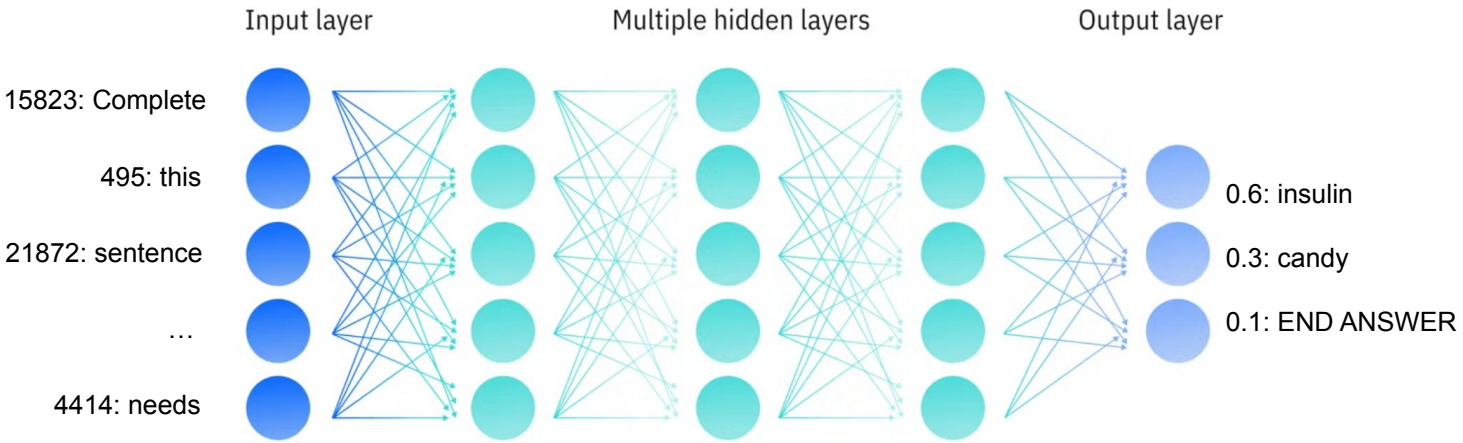
Tokens	Characters
11	63

> 💡 LLMs charge by the number of tokens in input **and** output — longer prompts = higher cost.

Complete this sentence: The patient with Type II Diabetes needs

[15823, 495, 21872, 25, 623, 10323, 483, 7003, 9839, 59910, 4414]

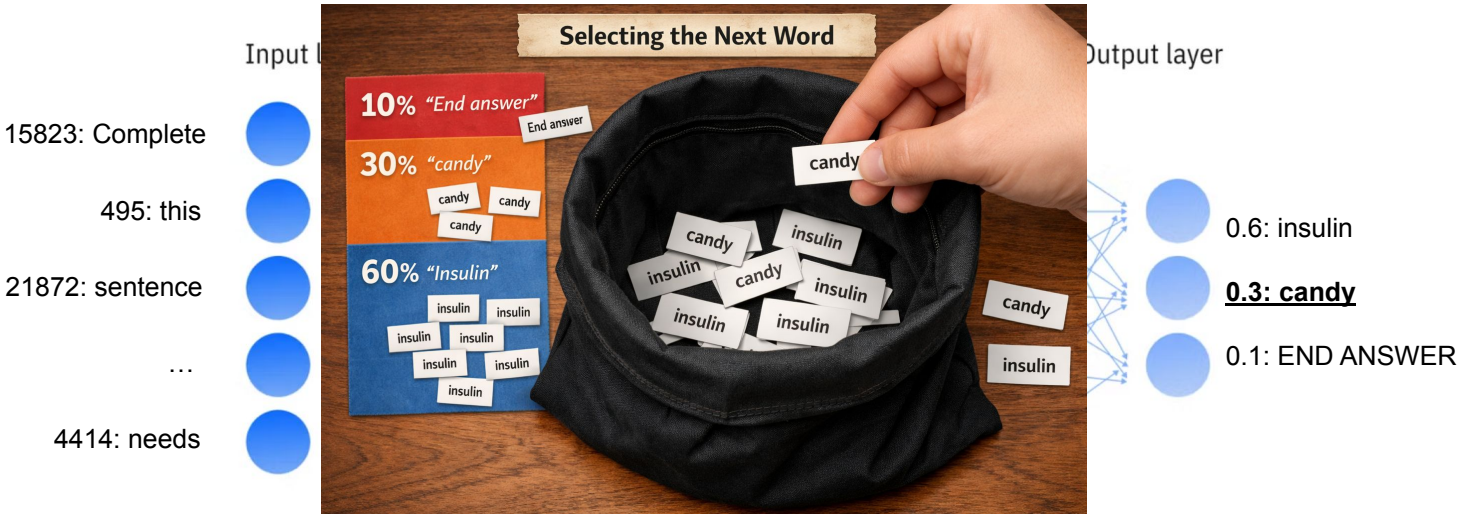
Step 2: Assign probabilities for every word



Input: “Complete this sentence: The patient with Type II Diabetes needs”

Output:

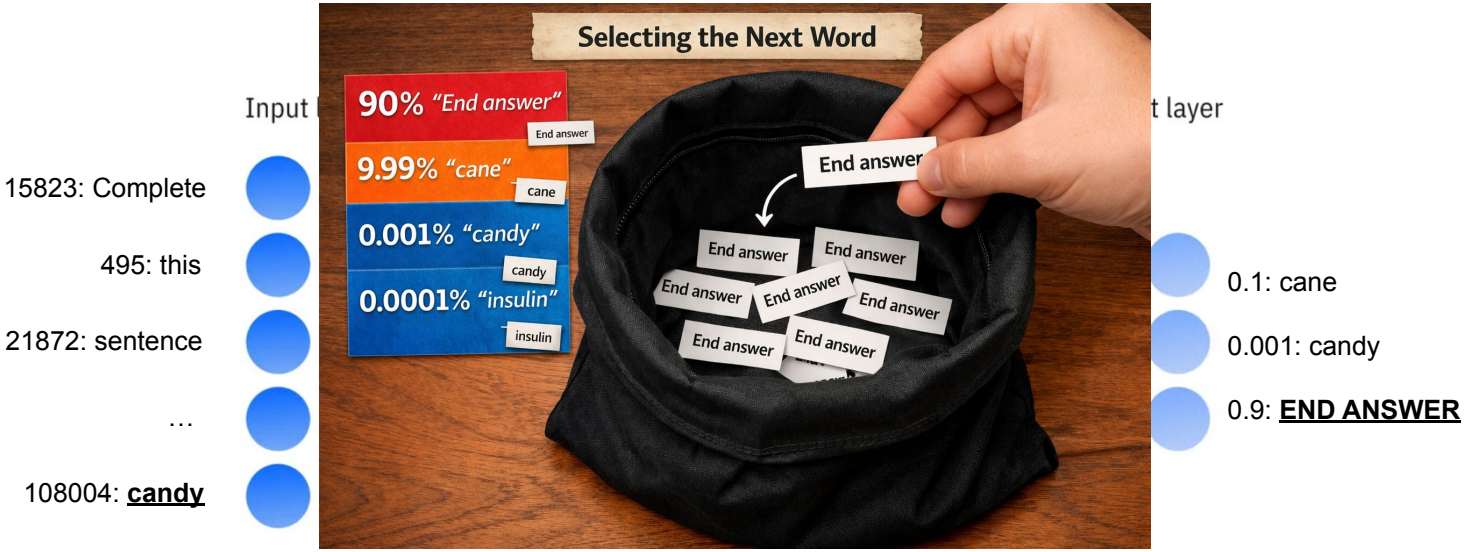
Step 2b: Picks a word based on those probabilities



Input: “Complete this sentence: The patient with Type II Diabetes needs”

Output: “**candy**”

Step 2c: Repeat until complete



Input: “Complete this sentence: The patient with Type II Diabetes needs candy”

Output: “END ANSWER”

Cumulative output: “**candy**”

How ChatGPT Really Answers

Response A — Concise

The patient with Type II diabetes needs **ongoing lifestyle management, regular monitoring of blood glucose, and individualized medical care to maintain safe glycemic control and reduce long-term complications**.

Response B — Detailed

The patient with Type II diabetes needs comprehensive lifestyle modification, regular blood glucose monitoring, individualized pharmacologic therapy (such as metformin or insulin when indicated), routine screening for complications, and ongoing education and support to effectively manage their condition and reduce the risk of cardiovascular, renal, neurologic, and ophthalmic complications.

Active Learning Check

Which of the following are true about how LLMs operate?

Select all that apply

1

LLMs only generate **1 token** of the answer at a time

2

LLMs do **not always** pick the most likely next token

3

Given the same prompt, an LLM will **always** produce the same output

4

An LLM will only perform **one round** of computation for a multi-token answer



Answer Revealed

✅ True

LLMs generate **1 token at a time** — each token is a separate output step

✅ True

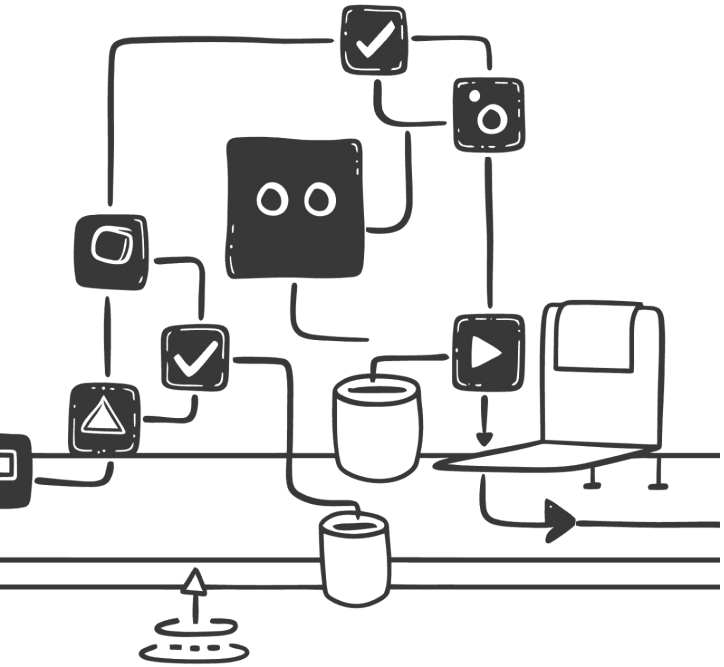
LLMs use **sampling** — they don't always pick the top token

❌ False

Same prompt ≠ same output — randomness is **built in by design**

❌ False

Multi-token answers require **many rounds** of computation — one per token



How Are LLM Models Developed?

1) Gather data
2) Train the model
3) Implement model into real world

Every LLM follows this fundamental pipeline — from raw data to real-world application.

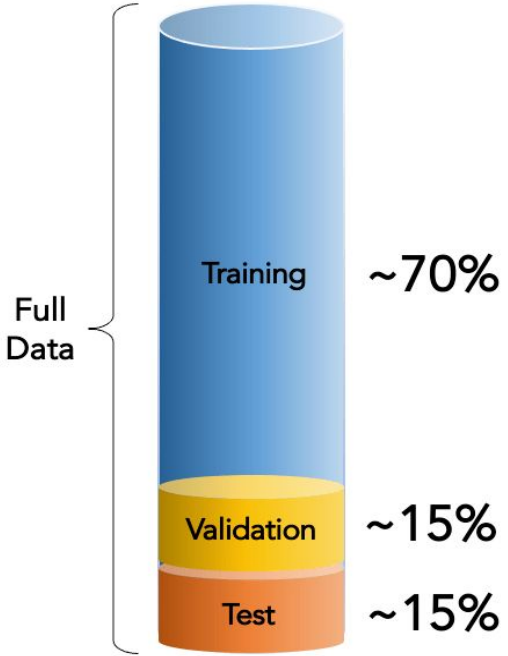
How does an LLM get made?

1) Gather data: real and synthetic









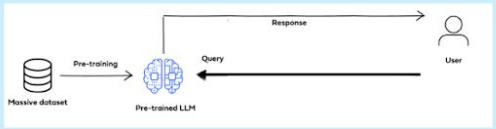
thaddeus-segura.com/train-test-split/

How does an LLM get made?

1) Gather data: real and synthetic
2) Training:
    a) Pretraining: read text → patterns are learned
    b) Supervised fine-tuning: good examples given
    c) +/- Reinforcement learning: improved with human feedback
    d) +/- Extra features like Retrieval Augmented Generation: an external source are added as context to give better answers (less hallucinations).

Retrieval Augmented Generation (RAG)

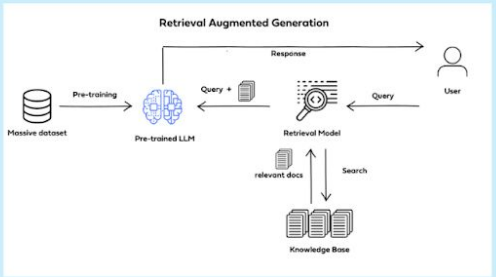
I am a physician seeing a patient with biochemically recurrent prostate cancer. I would like to discuss the EMBARK clinical trial with a trainee who is seeing the patient with me. Please highlight the patients included in the trial, the intervention done and efficacy endpoints



vs

NEJMoa2303974.pdf
PDF

I am a physician seeing a patient with biochemically recurrent prostate cancer. I would like to discuss the attached trial with a trainee who is seeing the patient with me. please summarize the attached article. Please highlight the patients included in the trial, the intervention done and efficacy endpoints



Step 6: Deploy & Monitor

The pipeline never truly ends



After deployment, **continuous oversight** is essential to ensure safety, accuracy, and fairness.

- **Monitor outputs**

  Track errors, hallucinations, and bias

- **Collect feedback**

  User reports and expert review

- **Retrain or patch**

  Update weights or add guardrails as needed

Post-implementation oversight: also known as AI governance.

Implementing AI Governance and Oversight



The University of Texas at Austin
Enterprise Technology

To ensure responsible use of AI platforms and tools, Enterprise Technology has convened a cross-organizational working group to:

- **Develop policies, standards, and guidelines** for the responsible and ethical use of AI tools and data, including a review of existing information, IT, and data governance policies as they relate to AI.
- **Create Training and Awareness Programs** to ensure the community is informed about responsible AI use and how to effectively use the approved AI tools.
- **Establish Feedback Mechanisms** to receive input and continuously improve the policies, guidelines, and programs.

https://tech.utexas.edu/governance/guidance-for-ai

Ethical Considerations

As LLMs become embedded in consequential decisions — in healthcare, law, education — **who is responsible** when they go wrong?



BLUEPRINT FOR AN

AI BILL OF RIGHTS

MAKING AUTOMATED SYSTEMS WORK FOR THE AMERICAN PEOPLE

OCTOBER 2022

THE WHITE HOUSE
WASHINGTON



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation



Human Alternatives, Consideration, and Fallback

AI Bill of Rights *Blueprint*



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation



Human Alternatives, Consideration, and Fallback

You should be protected from unsafe or ineffective systems.

...proactively...mitigate potential harms

...independent evaluation and reporting...

https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/

AI Bill of Rights *Blueprint*



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation



Human Alternatives, Consideration, and Fallback

You should not face discrimination by algorithms and systems should be used and designed in an equitable way.
...proactive equity assessments...
...use of representative data..
...ensuring accessibility..

https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/

AI Bill of Rights *Blueprint*



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation



Human Alternatives, Consideration, and Fallback

You should be protected from abusive data practices via built-in protections and you should have agency over how data about you is used.

...protections are included by default...
...strictly necessary data...

https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/

AI Bill of Rights *Blueprint*



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation



Human Alternatives, Consideration, and Fallback

You should know that an automated system is being used and understand how and why it contributes to outcomes that impact you.

https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/

AI Bill of Rights *Blueprint*



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation

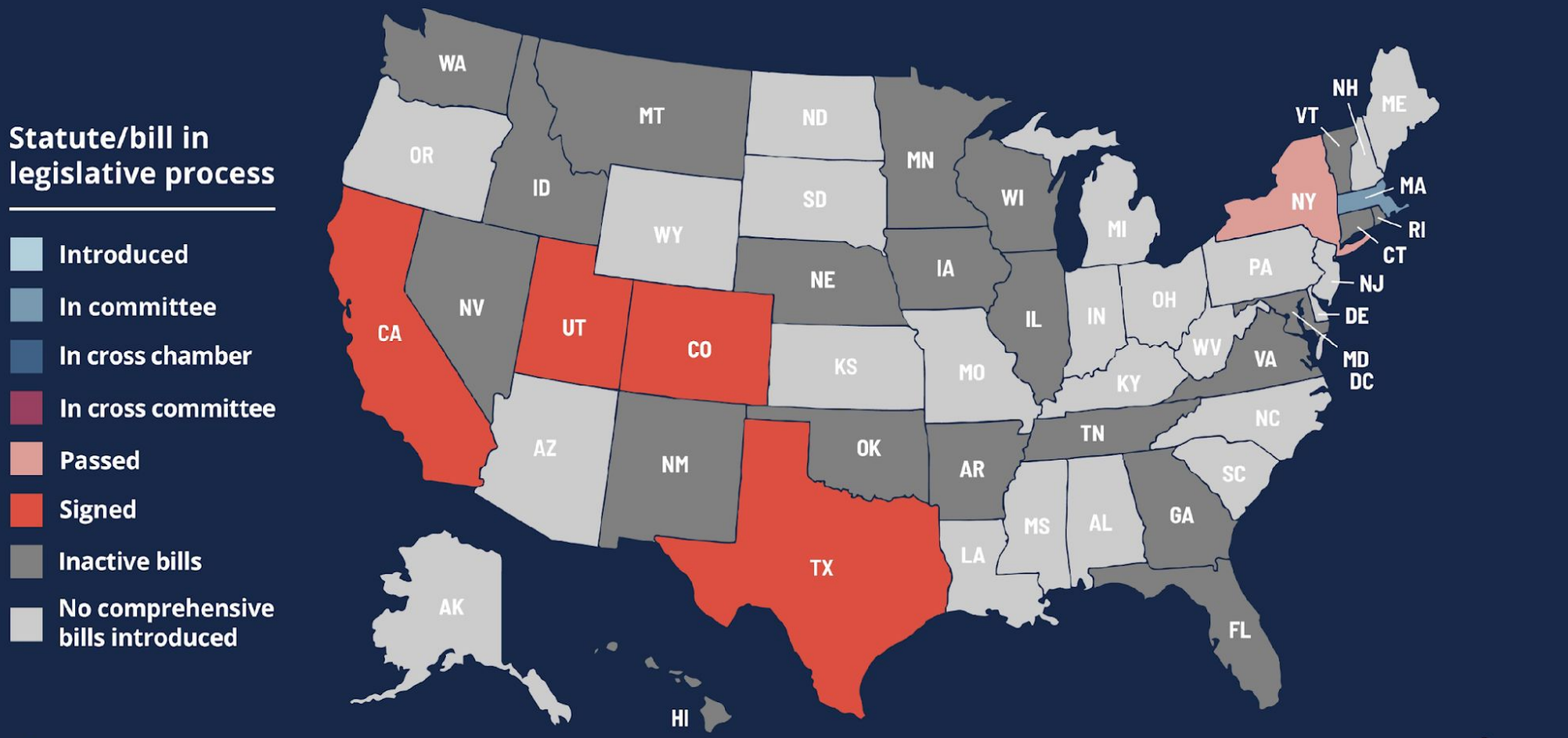


Human Alternatives, Consideration, and Fallback

You should be able to opt out, where appropriate, and have access to a person who can quickly consider and remedy problems you encounter.

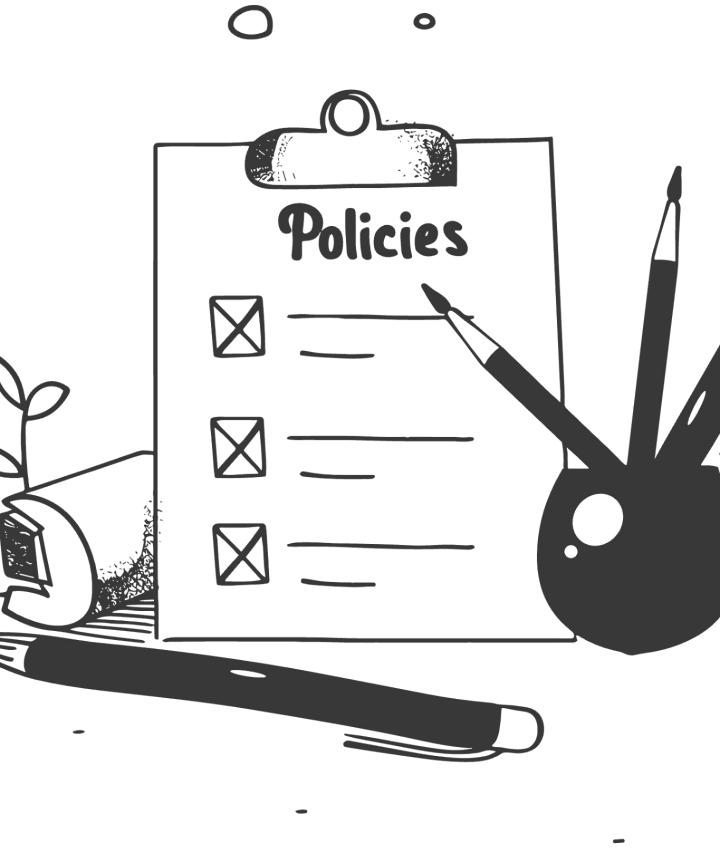
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US State AI Governance Legislation Tracker 2025



Last updated 6 October. 2025

iap



AI Governance: What's Happening in Texas

TRAIGA (HB 149)

Texas Responsible AI Governance Act — effective **January 2026**

SB 1188

AI may assist care — **clinicians remain accountable**

State Oversight

AI council + enforcement by **Attorney General**

→ **Transparency**

Disclose AI use in high-stakes decisions

→ **Safety**

Mitigate algorithmic harm across populations

→ **Clinician Accountability**

Human oversight remains non-negotiable



Liability and Oversight

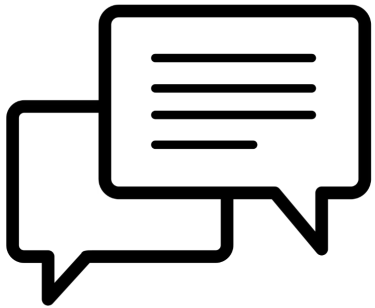
Your license. Your responsibility.



Stay current on local guidelines — institutional, state, and federal — around ethical AI use. **Ultimately, you are accountable for every clinical decision.**

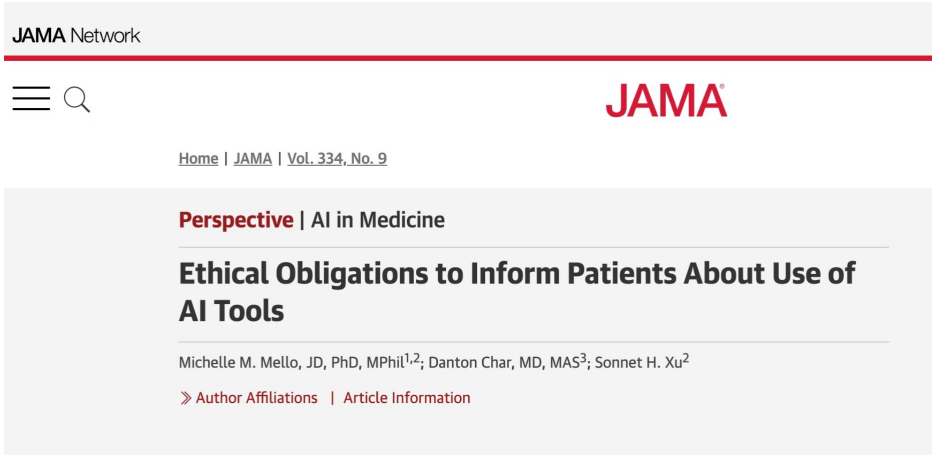
DISCUSSION QUESTION

How and when do you disclose AI use to patients?



JAMA PERSPECTIVE

Ethical Obligations to Inform Patients About AI





Michelle M. Mello, JD, PhD, MPhil

Michelle M. Mello, JD, PhD, MPhil — Stanford Law School

Published in *JAMA*, Vol. 334, No. 9

Argues that clinicians have a duty to disclose AI tool use when it materially affects patient care decisions or outcomes.

FRAMEWORK

AI Use Cases: Consent, Notification, or Neither?

Table. Illustrative Artificial Intelligence (AI) Use Cases and Recommendations for Patient Notification

Use case	Recommendation and rationale[a]
Consent	
AI-guided, nonautonomous surgical robot	Consent, because surgery typically involves considerable risk of harm and patients can choose nonrobotic surgery.
Tool that analyzes the genomic information of patients with cancer to predict responses to specific drugs and recommend treatment plans	Consent, because inaccuracy could lead to poor outcomes and patients can opt out of having the tool used.
Notification	
Predictive algorithm to identify patients who may have hypertrophic cardiomyopathy based on existing echocardiography data	Notification of those who screen positive, because although patients cannot avoid having the tool used, knowing that follow-up echocardiography is being recommended because of an AI algorithm may affect their decision about whether to accept the recommendation.
Generative AI tool that drafts replies to patient emails	Notification, because inaccuracy may cause harm and informed patients may be activated to question odd email replies.
Ambient listening tool that generates clinic visit summaries	Notification, because inaccuracy may cause harm and patients can check summaries for accuracy. (In some states, however, consent is legally required.)
Neither consent nor notification	
Predictive algorithm to inform decision whether to stock blood in operating room for potential transfusion	Neither, because patients cannot influence such operational decisions and ensuring blood availability does not replace patient consent for blood transfusion.
AI-assisted mammography interpretation	Neither, assuming AI-assisted interpretation outperforms human only, because exercising the choice to seek alternate care elsewhere would increase risk of harm.
Generative AI tool that drafts summaries of imaging findings dictated by radiologist	Neither, because risk of harm is low given that tool only summarizes radiologist's input.
Predictive algorithm for identifying inpatients for whom daily laboratory testing can be safely discontinued	Neither, because patients cannot opt out of use of the tool or demand that laboratory tests be continued; and because the determination to order laboratory tests comes from the clinician and the alert merely suggests laboratory tests might be safely discontinued.
Generative AI tool to help hospital file prior authorization requests with insurers	Neither, because risk of harm is low (given that denials can be appealed) and patients cannot influence such operational decisions or personally review prior authorization requests for accuracy.

[a] Recommendations assume that the AI model has performed well in predeployment testing on a similar population.

SAMPLE LANGUAGE

Consent in Practice: Ambient Scribe Example

Apex Hospital's model: written information sheet at check-in + oral consent before the exam room. Physicians review and edit all AI-generated summaries before finalization.

Box. Sample Patient Consent for an Ambient Scribe Tool for Clinic Visits[a]

Modality

Information is distributed via paper information sheet when patient checks in for clinic visit; nurse elicits oral consent when patients are brought to examination room.

Information Sheet

Apex Hospital uses an artificial intelligence (AI) tool called Intelligent Ear to help your physician take notes on your visit today and summarize them in your medical record. Under California law, you have the right not to have this tool used.

Intelligent Ear makes an audio recording of your physician visit and uses AI to write out the conversation in full. It then uses a different kind of AI to generate a summary of the important parts of the conversation. Your physician will have the opportunity to review and edit this summary before making it part of your medical record.

The audio recording is securely sent to the company that developed Intelligent Ear, but the company does not keep any information about you once the visit summary is created.

Apex Hospital is using this tool because of evidence that it can improve care. Having this type of aid enables physicians to keep their focus on you during your visit and save time writing notes at the end of the day. Testing showed Intelligent Ear generates good-quality summaries and that physicians find it helpful.

The tool performs somewhat less well for people with accented English, people with speech impediments, and conversations with more than 2 speakers. To help avoid inaccuracies, your physician has been trained to review summaries with special care in those circumstances. Apex Hospital is using this tool after studying the results of a pilot test at Apex, and continues to monitor the tool's performance to make sure it is working well for all types of patients.

We are excited to be able to offer this tool in our continuing efforts to deliver the best care possible. If you do not wish to have the tool used in your visit today, please let your nurse know.

Nurse Script

Did you have any questions about the scribe tool that the physician uses? Is it OK with you if the physician uses it today?

[a] For use in a state where 2-party consent for recording conversations is legally required. In other states, notification without nurse confirmation of consent would be ethically sufficient.



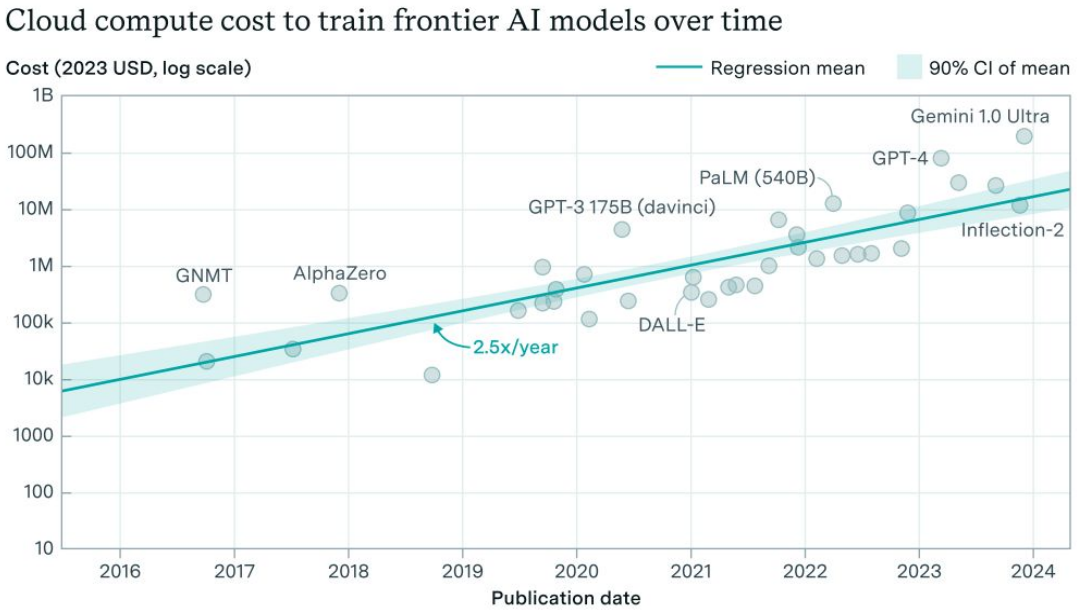
Cost & Environmental Impact

AI is not free — financially or ecologically. Understanding these trade-offs is essential for **responsible adoption** in healthcare settings.

Financial Cost

Energy Use

Water Footprint



COST TRAJECTORY

Training Costs Are Rising Exponentially

Cloud compute cost to train frontier AI models has grown **~2.5× per year** since 2016 (epoch.ai). AI companies are not currently profitable — **expect prices to rise.**



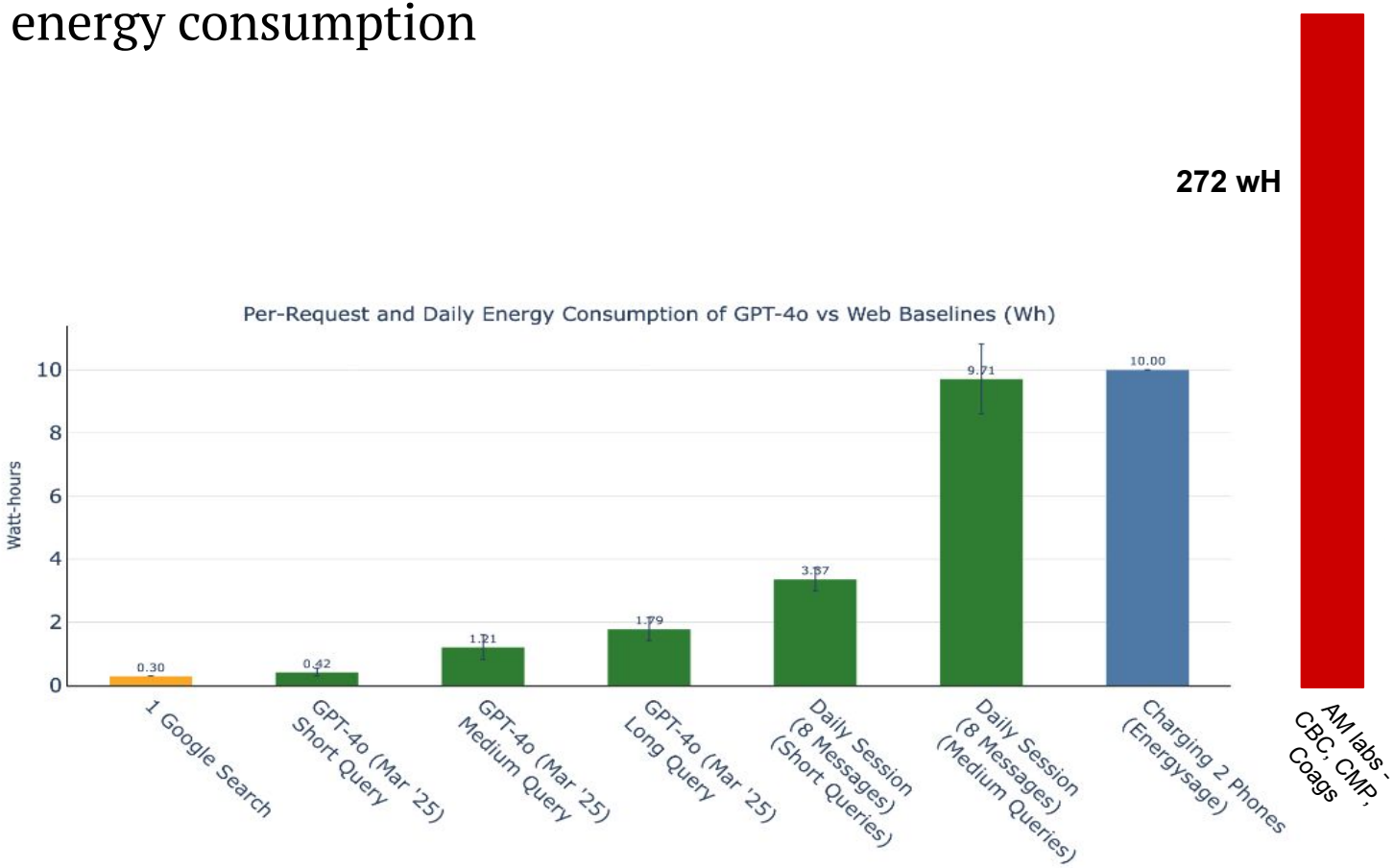
KEY CONCEPT

AI tools should add *value*.

Many clinical AI tools have **not yet demonstrated** meaningful benefit in rigorous trials. Before adopting a tool, ask:

1. Does evidence support this use case?
2. Does it improve outcomes — or just add complexity?
3. Is the cost (financial + environmental) justified?

AI's energy consumption

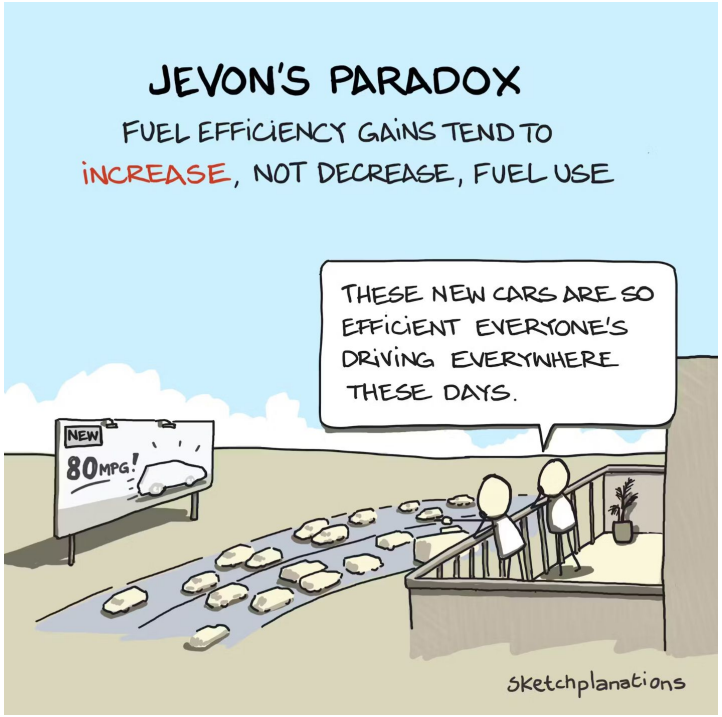


Jeghem et al. *How Hungry is AI?*

Spoyalo et al. *Patient, hospital and environmental costs of unnecessary bloodwork*

CONCEPT TO WATCH

Jevon's Paradox & AI



Efficiency gains can increase total consumption.

As AI tools become more efficient, usage — and total resource consumption — may rise dramatically.

In healthcare: more efficient AI documentation may lead to **more queries, more tools, more data** — not less energy use.

Efficiency ≠ sustainability. Monitor adoption curves alongside environmental metrics.

Course Outline

Session 1: Introduction to AI

What is AI? Why does it matter for healthcare?

Session 2: Error & Bias in AI Systems

AI for clinical decision-making — understanding failure modes

Session 3: Evaluating AI Systems

Benchmarks, critical appraisal frameworks, and validation

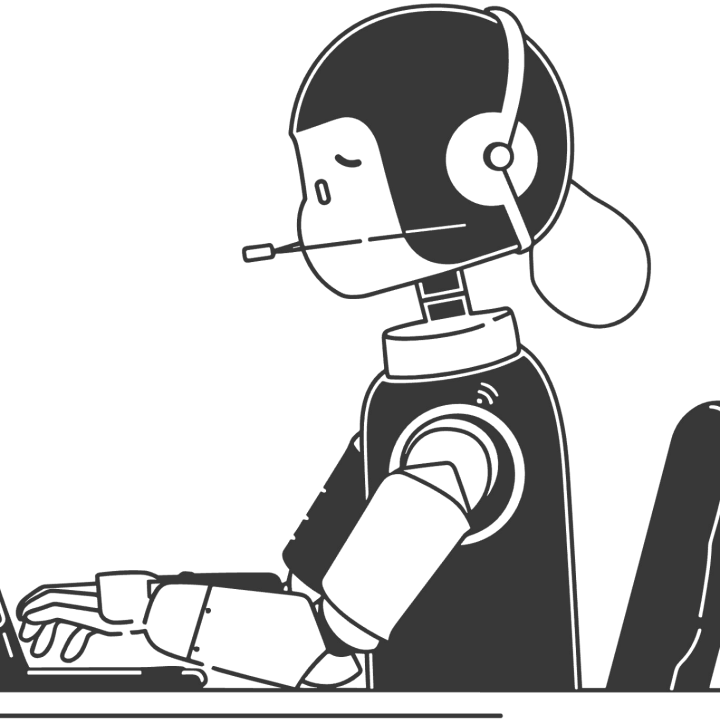
Session 4: Clinical Applications

AI for documentation, critical appraisal, and future directions



Niharika's AI: [Resource Guide]





🤖 Outsource This to AI

Stop spending hours on tasks a smart tool can handle in minutes. Here are three AI tools worth adding to your workflow — starting today.

01

Presentations

Gamma AI — slides from ideas in minutes

02

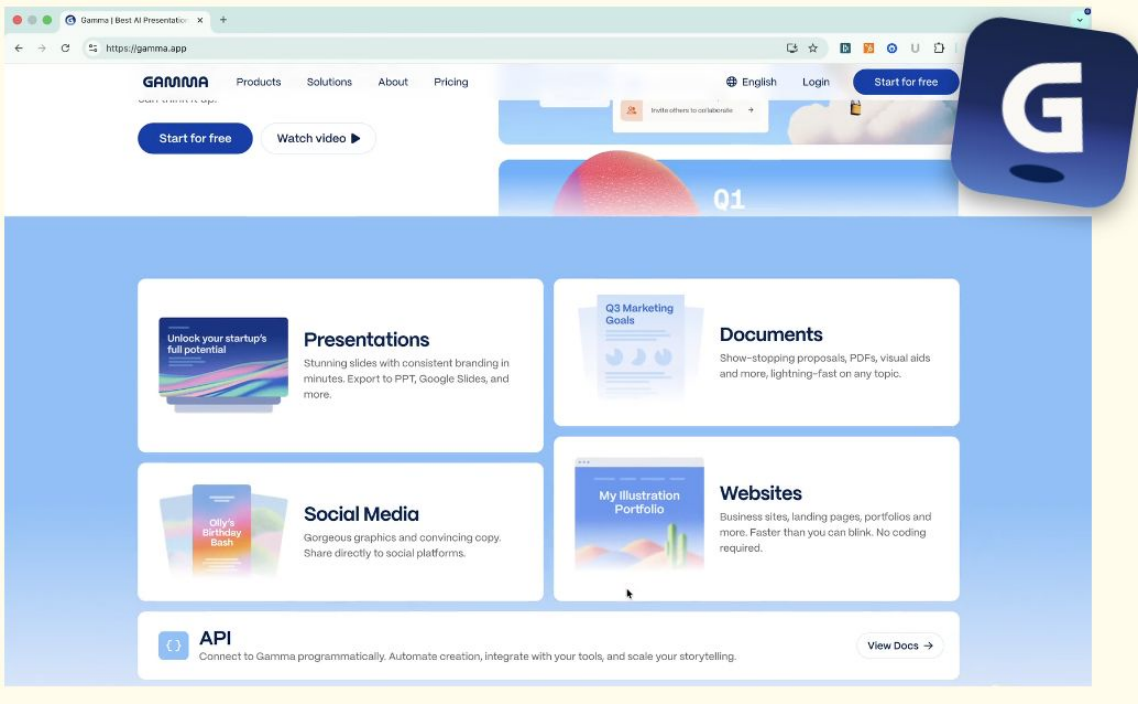
Literature Review

Elicit — finds and reads papers for you;

03

Professional Photos

InstaHeadshots — studio quality, no photographer



TOOL #1

🎨 Gamma AI

Presentations in minutes

- Turn rough ideas → **polished slides** instantly
- Perfect for lectures, quick decks, and teaching
- Replaces PowerPoint's designer — and then some

You're looking at Gamma right now



🔍 Elicit: Literature Review Made Effortless

Find Papers

Surfaces relevant studies instantly — no manual PubMed hunting

Read for You

Extracts key findings, methods, and limitations automatically

Perfect For

Journal club prep, research ideas, quick evidence review

TOOL #2

Think of it as...

> PubMed + an assistant that **actually reads** the papers

Save hours of literature review. Elicit does the heavy lifting so you can focus on insights.

Elicit Systematic Literature Review





TOOL #3

📸

InstaHeadshots

Studio-quality photos — no photoshoot needed

> Think: **"AI photographer in your pocket"**

→ Upload a few selfies → get **professional headshots**

→ Ideal for faculty profiles, LinkedIn, and conferences

→ No scheduling, no studio, no hassle